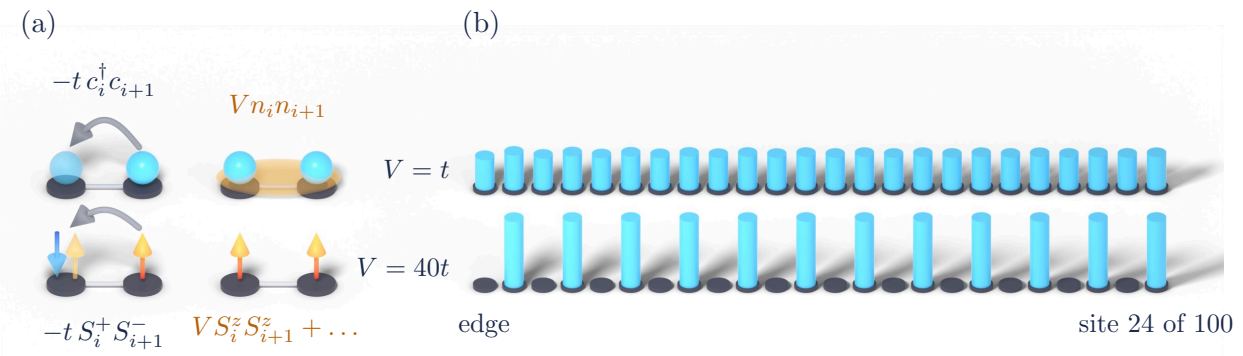

(a)
$-t\, c_i^\dagger c_{i+1}$
$V n_i n_{i+1}$
$-t\, S_i^+ S_{i+1}^-$
$V S_i^z S_{i+1}^z + \ldots$
(b)
$V = t$
$V = 40t$
edge
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